Suppose $E(x) = K$ for $x \in B_R(p)$

$E \geq K/2$ on $B_{K/4}(x)$

Can assume $p \notin B_{K/4}(x)$

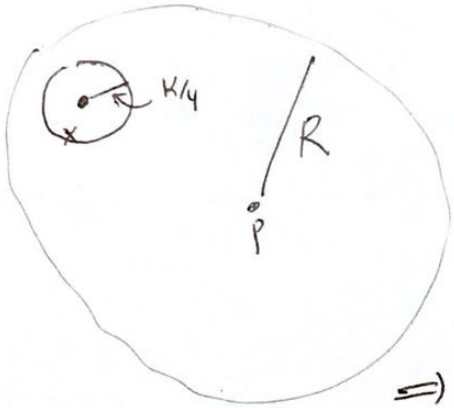
Let $c_K > 0$ s.t. $c_K \, \underline{L}_{2R}(K/4) = K/2$

$c_K \to 0 \iff K \to 0$ (using $\underline{M}^n_{-1}$)

$\underline{L}_{2R}$ :

$\Rightarrow \Delta\left(c_K \, \underline{L}_{2R}(\rho_x)\right) \geq c_K \geq \mathcal{F} E$

if $\varepsilon, L^{-1}, \delta$ small.




$c_K \, \underline{L}_{2R}(\rho_x) \leq E$

on $\partial\left(B_{2R}(x) \setminus B_{K/4}(x)\right)$

$\Rightarrow c_K \, \underline{L}_{2R}(\rho_x) \leq E$ on $B_{2R}(x) \setminus B_{K/4}(x)$

So $\varepsilon \geq E(p) \geq c_K \, \underline{L}_{2R}(R)$

$\Rightarrow K \leq \mathcal{F}(\varepsilon, L^{-1}, \delta \mid n, R)$ $\square$

<u>Lemma</u> If $(*)$ holds then

$$|h_\pm - b_\pm| \le \Psi \quad \text{on } B_R(p)$$

<u>Proof</u> Laplacian comp: $\Delta b_\pm \le \Psi$

i.e. $\Delta(b_\pm - h_\pm) \le \Psi$

quantitative max principle $\Rightarrow$ $b_\pm - h_\pm \geq -\Psi$

Abresch - Gromoll $\Rightarrow$ $b_+ + b_- \le \Psi$

$$\Rightarrow \quad h_+ + h_- = (h_+ - b_+) + (h_- - b_-) + (b_+ + b_-) \le \Psi$$

$\forall x$, $b_+(x) + b_-(x) = d(x, q_+) - d(p, q_+) + d(x, q_-) - d(p, q_-)$

$$= d(x, q_+) + d(x, q_-) - d(q_-, q_+) - \big(d(p, q_+) + d(p, q_-) - d(q_-, q_+)\big)$$

$$= E(x) - E(p) \geq -\varepsilon$$

$\Delta(h_+ + h_-) = 0$ so by max principle $h_+ + h_- \geq -\varepsilon$

Finally $-\Psi \le b_+ - h_+ = b_+ + b_- - (h_+ + h_-) - (b_- - h_-)$

$$\le \Psi + \varepsilon + \Psi$$

$\square$

<u>Lemma</u> Assuming $(*)$

$$\fint_{B_R(p)} |\nabla b_+ - \nabla h_+|^2 \le \Psi \qquad \left( \fint_U = \frac{1}{\mathrm{Vol}(U)} \int_U \right)$$

<u>Proof</u>

$$\fint_{B_R(p)} |\nabla b_+ - \nabla h_+|^2 = - \fint_{B_R(p)} (b_+ - h_+) \Delta b_+$$

signed measure

$$\le \Psi \cdot \fint_{B_R(p)} |\Delta b_+|$$

Laplacian comp $\Rightarrow$ $\Delta b_+ \le \Psi$ on $B_R$ as distributions.

$$\left| \int_{B_R(p)} \Delta b_+ \right| = \left| \int_{\partial B_R(p)} \nabla_n b_+ \right| \le \mathrm{Vol}(\partial B_R(p))$$

$$\Rightarrow \quad \left| \int_{B_R(p)} \underbrace{(\Delta b_+)^+ + (\Delta b_+)^-}_{\Delta b_+} \right| \le \mathrm{Vol}(\partial B_R(p))$$

$$\Rightarrow \quad \int_{B_R(p)} |(\Delta b_+)^-| \le \mathrm{Vol}(\partial B_R(p)) + \Psi \cdot \mathrm{Vol}(B_R(p))$$

$$\Rightarrow \quad \fint_{B_R(p)} |\Delta b_+| \le 2\Psi + \frac{\mathrm{Vol}(\partial B_R(p))}{\mathrm{Vol}(B_R(p))}$$

$$\le \frac{\mathrm{Vol}(\partial B_R(\underline{0}))}{\mathrm{Vol}(B_R(\underline{0}))} \text{ in } \underline{M}^n_{-1}$$

So $\displaystyle\fint_{B_R(p)} |\nabla b_+ - \nabla h_+|^2 \le \Psi$ $\square$

<u>Lemma</u> Assuming $(*)$

$$\fint_{B_{R/2}(p)} |\mathrm{Hess}_{h_+}|^2 \le \Psi$$

<u>Proof</u> Bochner: $\frac{1}{2}\Delta\left(|\nabla h_+|^2 - 1\right) = |\mathrm{Hess}_{h_+}|^2 + \underbrace{\mathrm{Ric}(\nabla h_+, \nabla h_+)}_{\ge -(n-1)\delta\,|\nabla h_+|^2}$

use cutoff fn. $\phi$ s.t. $\phi = 1$ on $B_{R/2}(p)$
$\mathrm{supp}(\phi) \subset B_R(p)$

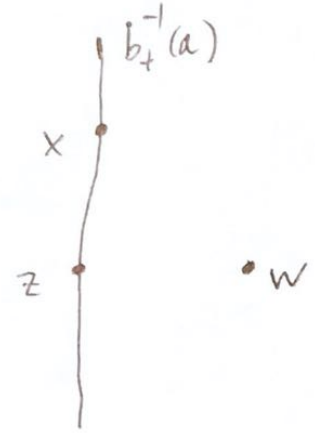
$$\fint_{B_R(p)} \phi\, |\mathrm{Hess}_{h_+}|^2 = \fint_{B_R(p)} \frac{1}{2}(\Delta\phi)\left(|\nabla h_+|^2 - 1\right) - \phi\,\mathrm{Ric}(\nabla h_+, \nabla h_+)$$

$$\le \fint_{B_R(p)} C(n,R)\,\left||\nabla h_+|^2 - 1\right| + (n-1)\delta\,|\nabla h_+|^2$$

$$\le \Psi$$

$\square$

<u>Lemma</u> Assume $(*)$. Let $x, z, w \in B_{R/8}(p)$ s.t. $x \in b_+^{-1}(a)$, $z \in b_+^{-1}(a)$ is a closest point to $x$ on $b_+^{-1}(a)$

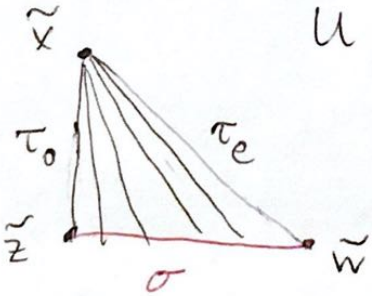



Then $\left| d(x,w)^2 - d(x,z)^2 - d(z,w)^2 \right| \leq \Psi$

<u>Proof</u> Apply iterated segment ineq. to $|\mathrm{Hess}_{h_+}|$

$\Longrightarrow$ can find $\tilde{x}, \tilde{z}, \tilde{w}$ s.t. $d(x,\tilde{x}), d(\tilde{z},z), d(w,\tilde{w}) < \Psi$

and if $\sigma : [0, \ell] \to M^n$ min. geodesic from $\tilde{z}$ to $\tilde{w}$ then for set $U \subset [0,\ell]$ of full measure for all $s \in U$ the min. geodesic $\tau_s : [0, \ell(s)] \to M^n$ joining $\tilde{x}$ to $\sigma(s)$ is unique

and

$$\int_U \int_0^{\ell(s)} |\mathrm{Hess}_{h_+}(\tau_s(t))| \, dt \, ds \leq \Psi$$